

Network Architecture

VLANs, firewalls, DMZ, and WAN design

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1. Introduction

This document describes the network architecture of ArchiSurance, including network segmentation, firewall policies, device inventory, and IP addressing. The Security Architecture (EA-SEC-006) covers the security aspects of the network, and the Infrastructure Design Document (EA-INFRA-003) provides the compute layer context.

Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The vendor selection process followed the procurement guidelines established by the CFO's office. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Security review was completed in accordance with the CISO's security-by-design framework. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel.

The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration

accounts for this burst pattern. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Change management activities include stakeholder communication, training sessions, and documentation updates.

Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request.

The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns.

2. Network Segmentation

Corporate LAN

The Corporate LAN segment provides connectivity for its assigned devices and nodes. Connected devices: PA Firewall, Network Switch Cisco Nexus, Wireless Access Controller.

Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The Corporate LAN has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Technical debt in this area was cataloged

during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Change management activities include stakeholder communication, training sessions, and documentation updates. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Security review was completed in accordance with the CISO's security-by-design framework. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity.

the DMZ

The Demilitarized Zone segment provides connectivity for its assigned devices and nodes. Connected devices: Load Balancer F5, Firewall Palo Alto.

The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Integration testing spans three environments: development, staging, and pre-production. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The DMZ Network has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings.

Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Lessons learned from the pilot phase have been incorporated into the current design iteration.

Cloud VPN

The Cloud VPN segment provides connectivity for its assigned devices and nodes. Connected devices: WAN Accelerator.

The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The vendor selection process followed the procurement guidelines established by the CFO's office. Integration testing spans three environments: development, staging, and pre-production. Dependencies on third-party services were documented and risk-assessed,

with fallback mechanisms defined for each critical integration point. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios.

Management VLAN

The Management VLAN segment provides connectivity for its assigned devices and nodes. Connected devices: Network Switch Cisco Nexus.

During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The Management VLAN has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Lessons learned from the pilot phase have been incorporated into the current design iteration. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. Change management activities include stakeholder communication, training sessions, and documentation updates. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple

producers from consumers and enable replay for debugging and reprocessing scenarios.

3. Firewall Rules

The Firewall Palo Alto enforces traffic policies between all network segments. The following table summarizes key rules:

Source	Destination	Port	Action
Internet	DMZ (LB F5)	443	Allow
DMZ	App Cluster	8080	Allow
App Cluster	DB Primary	5432	Allow
App Cluster	DB Primary	1521	Allow
Mgmt VLAN	All Nodes	22	Allow
Corporate LAN	Internet	443,80	Allow
Any	Any	Any	Deny

The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Stakeholders across multiple departments have contributed requirements for this workstream. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The firewall rules policies has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap.

An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The incident management process defines four severity levels, with Severity 1 incidents

requiring a response within fifteen minutes and a post-incident review within five business days. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Stakeholders across multiple departments have contributed requirements for this workstream.

Change management activities include stakeholder communication, training sessions, and documentation updates. Lessons learned from the pilot phase have been incorporated into the current design iteration. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility.

An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The firewall policies rules management has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023.

4. Network Devices

the load balancer

Connects to: Demilitarized Zone.

The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA.

Firewall Palo Alto

Connects to: the DMZ, Office LAN.

Security review was completed in accordance with the CISO's security-by-design framework. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the

normal rate; the auto-scaling configuration accounts for this burst pattern. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns.

SAN Storage Array

Connects to: Database Server Primary, Database Server Replica.

The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

Backup Tape Library

Connects to: DR Recovery Node.

A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model.

HSM Appliance

The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel.

WAN Accelerator

Connects to: Cloud VPN.

Security review was completed in accordance with the CISO's security-by-design framework. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Change management activities include stakeholder communication, training sessions, and documentation updates. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

Network Switch Cisco Nexus

Connects to: Internal Network, Management VLAN.

Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. From a historical perspective, ArchiSurance first recognized the need for modernization in this

area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios.

Wireless Access Controller

Connects to: Corporate LAN.

Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Lessons learned from the pilot phase have been incorporated into the current design iteration.

5. IP Addressing Scheme

Segment	Subnet	Gateway	VLAN ID
Corporate LAN	10.10.0.0/16	10.10.0.1	100
DMZ Network	10.20.0.0/24	10.20.0.1	200
Management VLAN	10.30.0.0/24	10.30.0.1	300
Cloud VPN	172.16.0.0/16	172.16.0.1	N/A
Database Subnet	10.10.10.0/24	10.10.10.1	110
App Cluster Subne	10.10.20.0/24	10.10.20.1	120

The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies.

The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The operational acceptance criteria include automated smoke tests, performance baselines,

runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point.

Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Integration testing spans three environments: development, staging, and pre-production. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The IP addressing subnet management has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023.

6. Appendices

References and Related Documents

Doc ID	Title	Version
EA-OVR-001	Enterprise Architecture Overview	3.0
EA-APP-002	Application Portfolio & Inventory	4.2
EA-INFRA-003	Infrastructure Design Document	2.5
EA-BIZ-004	CRM Business Process Description	2.0
EA-CLM-005	Claims Processing Workflow	3.1
EA-SEC-006	Security Architecture	2.3
EA-INT-007	Integration Architecture	2.8
EA-DAT-008	Data Architecture & Governance	2.0
EA-CLD-009	Cloud Migration Assessment	1.5
EA-DR-010	Disaster Recovery & Business Contin	3.0
EA-NET-011	Network Architecture	2.2
EA-ROAD-012	Strategic IT Roadmap 2025-2027	2.0
EA-SVC-013	IT Service Catalog	3.5
EA-VND-014	Vendor & Technology Assessment	1.8
EA-OPS-015	Operational Runbook: Core Platform	4.1
EA-CMP-016	Compliance & Regulatory Requirement	2.5
EA-CHG-017	Change Management: ERP Migration	1.2

Distribution List

Name	Role	Responsibility
J. van den Berg	Enterprise Architect	Review & Approve
M. de Vries	Application Architect	Review & Contribute
P. Jansen	Integration Architect	Review & Contribute
K. Bakker	Infrastructure Architect	Review & Contribute
S. Visser	Business Analyst	Inform
T. Mulder	Process Architect	Inform
R. de Jong	CISO	Review & Approve
L. Hendriks	Data Architect	Inform
A. Willems	CTO	Final Approval
B. Smit	Head of Claims	Business Sign-off

Glossary of Acronyms

Acronym	Definition
API	Application Programming Interface
BPM	Business Process Management
CDN	Content Delivery Network
CISO	Chief Information Security Officer
CRM	Customer Relationship Management
DR	Disaster Recovery
DMS	Document Management System
DMZ	Demilitarized Zone
EA	Enterprise Architecture
ESB	Enterprise Service Bus
ETL	Extract, Transform, Load

FNOL	First Notice of Loss
GDPR	General Data Protection Regulation
HSM	Hardware Security Module
IAM	Identity and Access Management
KPI	Key Performance Indicator
LAN	Local Area Network
MTTR	Mean Time to Recovery
NPS	Net Promoter Score
PAS	Policy Administration System
PCI-DSS	Payment Card Industry Data Security Standard
PoC	Proof of Concept
RBAC	Role-Based Access Control
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SAN	Storage Area Network
SLA	Service Level Agreement
SOX	Sarbanes-Oxley Act
STP	Straight-Through Processing
VLAN	Virtual Local Area Network
VPN	Virtual Private Network
WAN	Wide Area Network

Appendix: Meeting Minutes - Architecture Review Board

Date: March 15, 2025. Location: ArchiSurance HQ, Room 4.12. Attendees: J. van den Berg (chair), M. de Vries, P. Jansen, K. Bakker, S. Visser, T. Mulder (remote). Apologies: R. de Jong (on leave).

Agenda item 1: Review of open action items from previous meeting. Action 2024-47 (Jansen): Provide updated cost estimate for Oracle migration. Status: Completed. The revised estimate is EUR 1.4M including contingency. Action 2024-48 (de Vries): Schedule penetration test with external vendor. Status: In progress, vendor shortlist reduced to two candidates.

Agenda item 2: Discussion on coffee machine replacement in the 3rd floor kitchen. The current Jura E8 has been experiencing frequent bean grinder failures. Facilities will procure a replacement by end of month. Not related to architecture but raised due to impact on team productivity during long review sessions.

Agenda item 3: Parking lot discussion about whether to adopt Backstage as the internal developer portal. Deferred to next meeting pending PoC results. K. Bakker noted that the onboarding documentation is outdated and should be refreshed regardless of the portal decision.

Next meeting: April 12, 2025 at 14:00 CET. Van den Berg to circulate agenda by April 5.

Detailed Revision Notes for Version 2.1

Section 2.3 was updated to reflect the new naming convention adopted by the Platform Engineering team. All references to 'microservice cluster' have been replaced with 'application workload group' per the Architecture Board decision of January 2025. Section 4.1 was expanded to include the results of the capacity planning exercise conducted in Q4 2024, which identified the need for additional compute resources in the Amsterdam primary data center.

Appendix B was added to include the RACI matrix requested by the PMO during the Q3 2024 quarterly review. The glossary was updated to include twelve new acronyms that were introduced since the previous version. Figure 3 was replaced with an updated diagram generated from the Sparx EA repository on March 28, 2025. The document classification was elevated from 'Internal' to 'Internal - Confidential' following the data classification review of February 2025.

Feedback from the peer review conducted by M. de Vries on March 10, 2025, addressed three items: (1) clarification of the boundary between the DMZ and the application tier, (2) addition of the backup retention policy reference, and (3)

correction of the SLA target for the Claims Handling Service from 99.5% to 99.9% as agreed in the contract amendment of December 2024.

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